

# Fraud Detection in Payment Systems

Fraud represents only a tiny fraction of transactions, and the patterns keep changing.

## The Massive Problem

# \$33B

projected global  
payment-card fraud annually

*Source: The Nilson Report, 2023-24*

## The Challenge

Identify unusual behavior **in real time**, across massive transaction volumes, without disrupting legitimate payments.

## Why AI/ML?

### Rules

Simple, but short-lived & easily bypassed

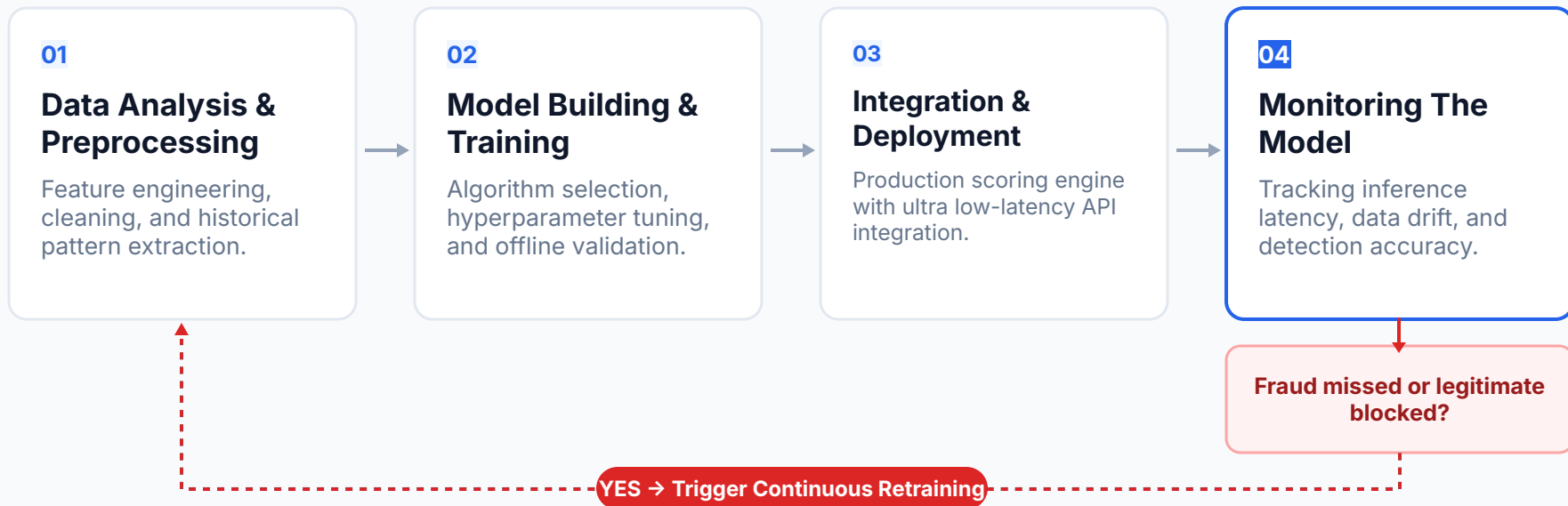
### Machine Learning

Learns complex, evolving patterns

**Goals:** Catch more fraud. Block fewer good transactions.

Do it in real time, at payment scale.

# ML-Based Fraud Detection Life Cycle



The most difficult part isn't building a good model — **it's running it efficiently and reliably at scale in production.**

# Credit Card Fraud Dataset

## Input Columns

Distance\_from\_home  
Distance\_from\_last\_transaction  
Ratio\_to\_median\_purchase\_price  
Repeat\_retailer  
Used\_chip  
Used\_pin\_number  
online\_order

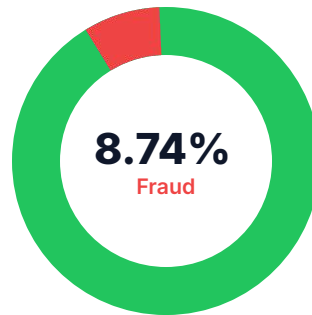
**Target Column: fraud**

## Total Transactions

**1,000,000**

Dataset Size  
1M Records

## Target Distribution



Non-Fraud Transactions

**912,597**

91.26%

Fraud Transactions

**87,403**

8.74%

# Out of the Box Models - Predictive AI

01

## Logistic Regression

Baseline statistical model, easy to start with and get basic understanding.

02

## Feed Forward Neural Network

Powerful and flexible classifier for detecting complex unknown patterns

Can do more advanced things, and increase model metrics.

**Model == Abstract Method**

**Training:** Discover fraud patterns in transaction history

**Inference:** Estimate probability that a given transaction is fraud

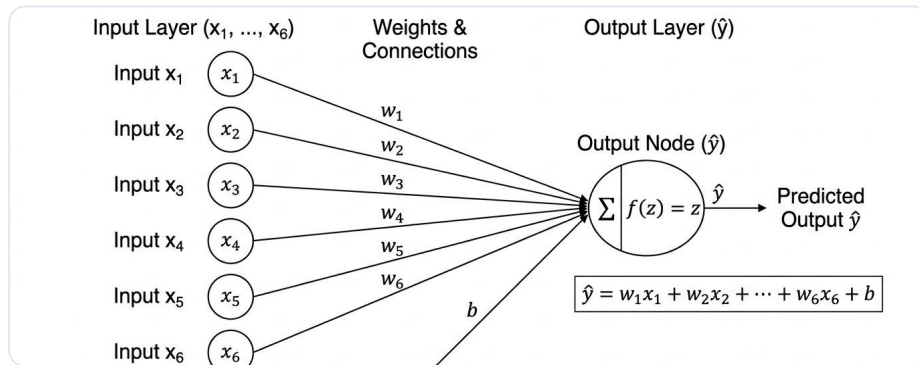
# Logistic Regression

- Simple binary classifier: true/false
- Outputs probability in range 0..1 that a transaction is fraud
- Easy to understand and explain
- Good start as a baseline
- Assigns weight to each input, can be interpreted

**Example:**

[CreditCardFraudLogisticRegression](#)

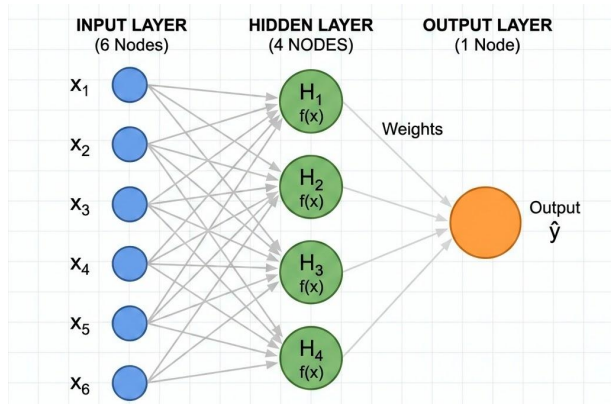
**Course:** [Logistic Regression](#)



```
FeedForwardNetwork logReg =  
FeedForwardNetwork.builder()  
    .addInputLayer(numInputs)  
    .addOutputLayer(1, SIGMOID)  
    .lossFunction(CROSS_ENTROPY)  
    .build();
```

# Feed Forward Neural Network Classifier

- Extends logistic regression with hidden layers
- Enables the model to deal with more complex patterns with higher accuracy
- Higher precision, less missed frauds



Example: [CreditCardFraudFFNetwork](#)

Course: [Neural Networks](#)

```
FeedForwardNetwork neuralNet =  
    FeedForwardNetwork.builder()  
        .addInputLayer(numInputs)  
        .addFullyConnectedLayer(32, RELU)  
        .addFullyConnectedLayer(16, RELU)  
        .addOutputLayer(1, SIGMOID)  
        .lossFunction(CROSS_ENTROPY)  
        .build();
```

```
neuralNet.getTrainer()  
    .setStopError(0.15f)  
    .setStopEpochs(10000)  
    .setLearningRate(0.001f);
```

# More Advanced Techniques

## 01. ENSEMBLES

### Hybrid Models

Ensemble / multiple models that complement existing solutions (Rules, Decision trees, etc.)

#### Key Highlights:

- Combines predictions across multiple model types
- Complements rule-based & decision tree logic
- Boosts accuracy and overall system robustness

## 02. GRAPH AI

### Graph Networks

Graph Attention Neural Networks for learning complex topological and relational structures.

#### Key Highlights:

- Graph Attention Neural Networks (GATs)
- Learns complex topological relationships
- Uncovers deep relational connectivity patterns